Ideation Phase Define the Problem Statements

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Team ID	6997f7cd05fb21b07dca8321
Project Name	Visualization Tool for Electric Vehicle Charge
Maximum Marks	

Customer Problem Statement

Topic: Visualization Tool for Electric Vehicle Charge

1. Target Customer

Electric vehicle owners and potential EV buyers who rely on electric vehicles for **daily commuting, long-distance travel, and sustainable transportation**.

2. Customer Needs

Customers need a **reliable, fast, and easily accessible charging system** so that they can travel without worrying about the vehicle running out of battery. They also need **accurate range information and efficient battery performance** to plan their trips confidently.

3. Customer Problems

Many EV users face several challenges related to charging and range, such as:

- **Limited driving range** compared to traditional fuel vehicles.
- **Long charging time**, especially with standard chargers.
- **Insufficient charging infrastructure**, particularly in rural or less developed areas.
- **Range anxiety**, where drivers worry that the battery will run out before reaching a charging station.

- **Difficulty locating available charging stations** during travel.

These problems create inconvenience and reduce confidence in using electric vehicles for long-distance travel.

4. Problem Statement

EV users struggle with **limited vehicle range, slow charging speed, and lack of easily accessible charging stations**, which leads to **range anxiety and inconvenience during travel**. Because of these challenges, many potential customers hesitate to adopt electric vehicles despite their environmental benefits.

5. Desired Outcome

Customers want a **faster charging system, improved battery range, and a reliable charging infrastructure** that allows them to travel long distances with confidence and convenience.

6. Opportunity

By improving **charging technology, battery efficiency, and charging station accessibility**, companies can create better EV solutions that enhance customer satisfaction and accelerate the adoption of electric vehicles.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An electric vehicle owner	charge my EV during long trips	it takes a long time to charge	fast charging stations are limited in many areas	**frustrated**
PS-2	An EV driver	travel long distances with my vehicle	the battery range is limited	charging infrastructure and battery capacity are not sufficient	worried / anxious